

ECON 422 Assignment 1

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1 Overview

The purpose of this assignment is to investigate how changes in the provincial minimum wage affect Ontarians, specifically looking into whether youth and youth-dominated industries are disproportionately affected. This will be done using data from the Labour Force Survey, which consists of monthly data for Ontario between 2016 and 2020. Based on this objective, and the reasons outlined below, my chosen sample period is 2017-2019.

2 Rationale

2.1 Employment Rate

Since the employment rate is the central measure in this analysis, it is worth establishing how it is defined first. Statistics Canada defines the employment rate as the number of employed individuals divided by the population aged 15 and over, expressed as a percentage (Government of Canada, 2017). When broken down by group (age, industry, full-time/part-time, and education in this case), the same ratio is applied using the employed count and population specific to that group. Employed, more specifically, refers to individuals who performed paid work or self-employment in the reference week. This definition also extends to unpaid family work, as well as individuals who held a job but were temporarily absent that week (Government of Canada, 2017).

2.2 Time Period

Ontario's minimum wage changed three times between 2016–2020: \$11.40 (Oct 1, 2016), \$11.60 (Oct 1, 2017), \$14.00 (Jan 1, 2018), and \$14.25 (Oct 1, 2020) (Government of Canada, n.d.). While the 2017 change was only a \$0.20 difference, the 2018 change of \$2.40 was much

larger, constituting a 21% increase relative to the previous 1.8% increase. Thus, when choosing the time frame, it was imperative to include at least these two policy changes, allowing the large 2018 change to be evaluated against the smaller 2017 change that preceded it. One consideration is that NAICS_21 (industry) data has no observations before 2017, which means that any subsetting involving industry, could only be constructed within the 2017–2020 time frame. This does not significantly distort the analysis since it still captures both the October 2017 and January 2018 minimum wage changes, which matters more than losing one year. Given this, the time frame needed to be lower bounded by 2017, and upper bounded in a way that captures 2018 in its entirety. 2019 presented an ideal selection for the upper bound since it satisfies this, but also avoids the arrival of COVID-19 in 2020. The pandemic depressed LFS sample sizes (294,963 observations in 2020, compared to 333,777–338,360 in 2017–2019) and triggered employment declines unrelated to minimum wage policy, making it unsuitable for isolating the effect of the 2018 increase.

2.3 Subsetting

To identify industries with a meaningful concentration of youth workers, I compared the youth share of employment across four selected industries. Youth workers (which I define as ages 15–24) account for 40.65% of employment in accommodation & food services and 26.01% in retail trade, compared to only 7.05% in health care & social assistance and 6.39% in professional, scientific & technical services. Given this gap, accommodation & food services and retail trade are the industries that should be observed for a minimum wage effect and are therefore used as the industry subsets. I include health care & social assistance and professional, scientific & technical services industry subsets as a comparison. I also decided to incorporate full and part-time status, as well as education level into my analysis, since they capture groups theoretically likely to be influenced by minimum wage changes.

3 Graphs

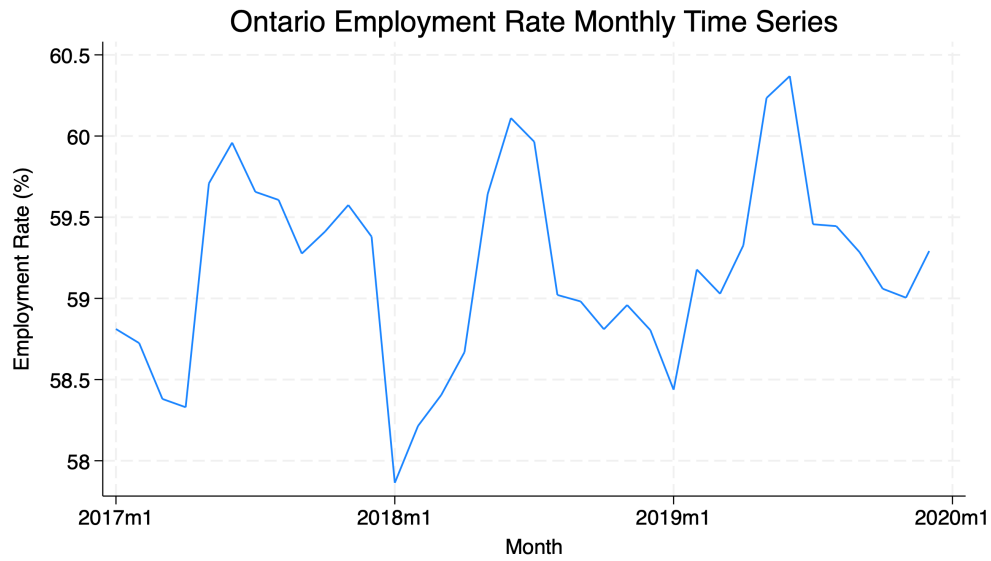


Figure 1: Employment Rate in Ontario from January 2017 to December 2019.

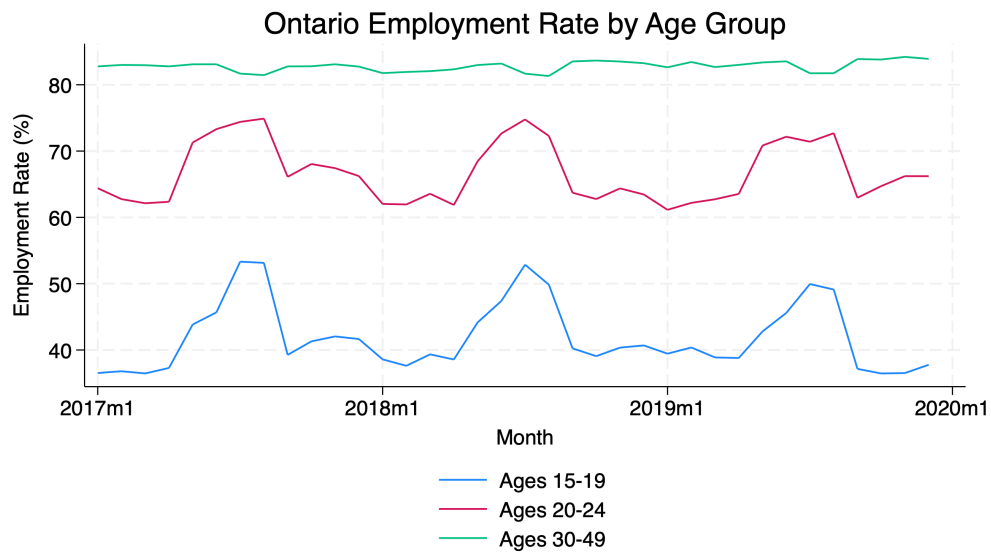


Figure 2: Employment Rate in Ontario from January 2017 to December 2019 by age group.

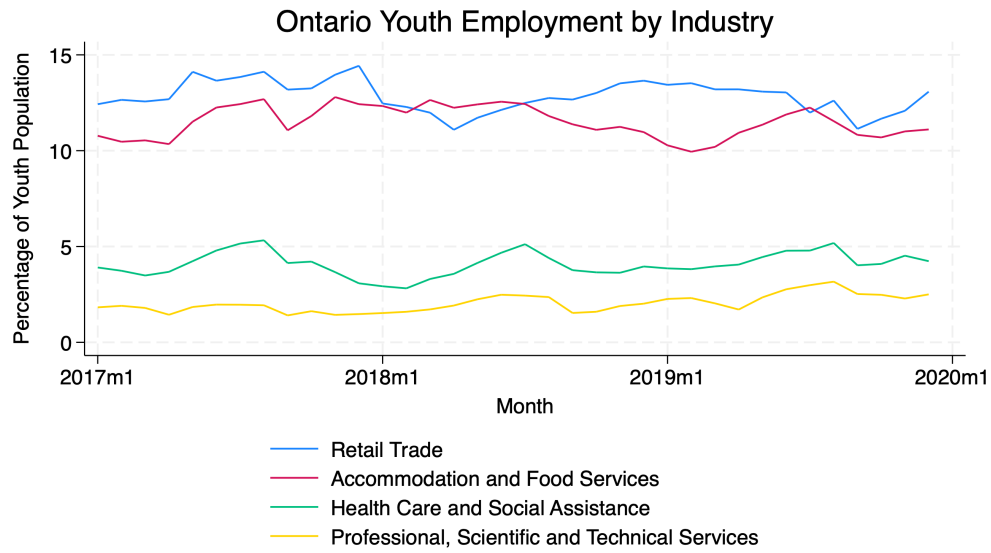


Figure 3: Employment Rate of Ontario youth (age 15-24) from January 2017 to December 2019 by industry.

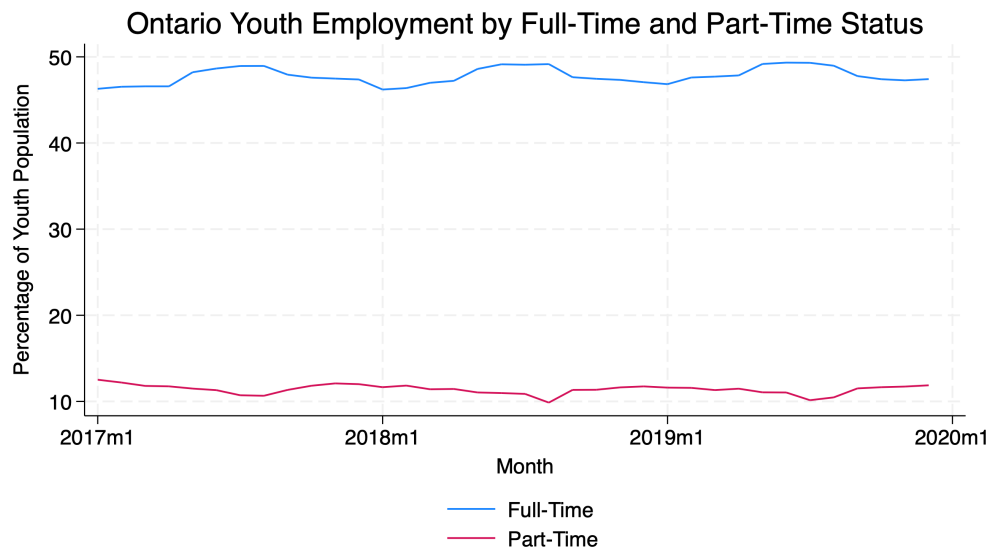


Figure 4: Employment Rate of Ontario youth (age 15-24) from January 2017 to December 2019 by full-time and part-time status.

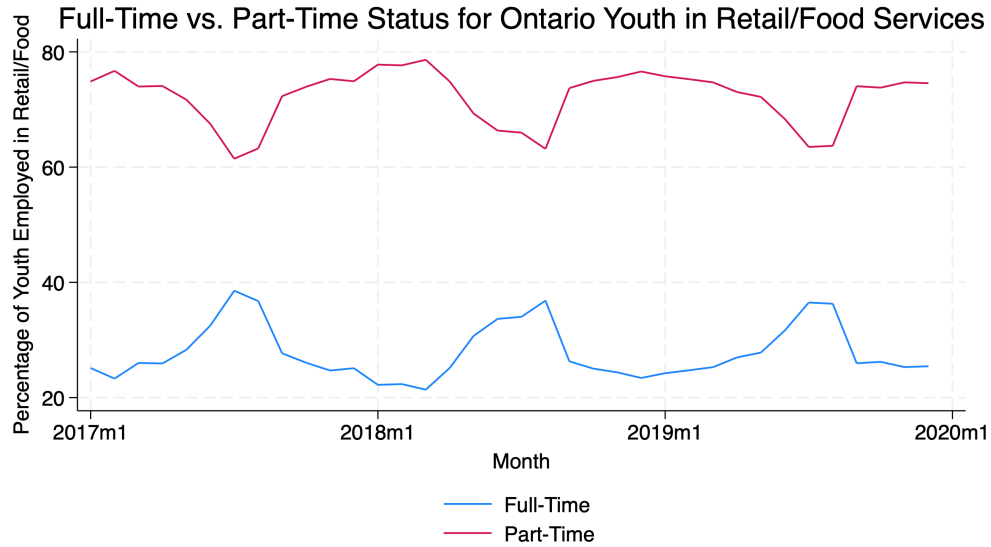


Figure 5: Employment Rate of Ontario youth (age 15-24) in retail/food sectors from January 2017 to December 2019 by full-time and part-time status.

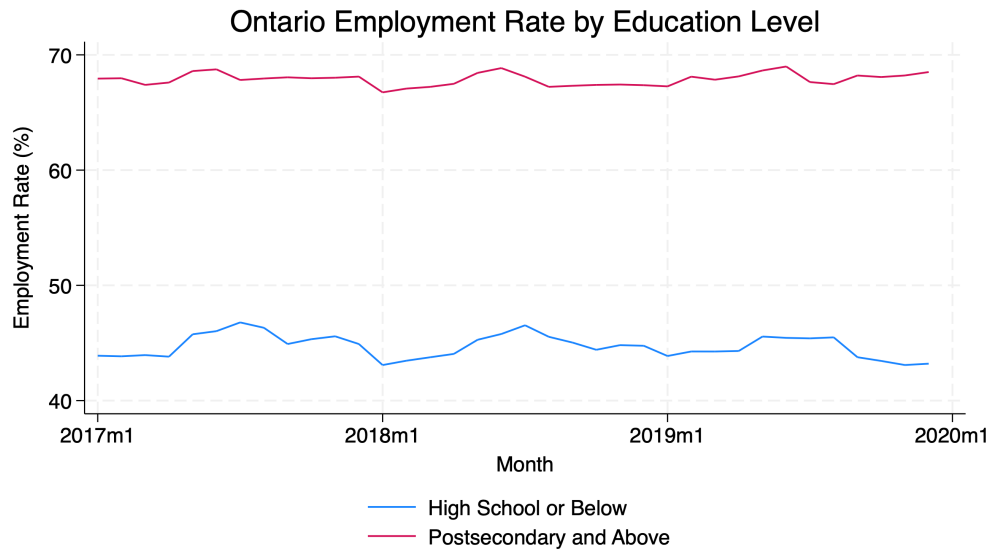


Figure 6: Employment Rate in Ontario from January 2017 to December 2019 by education level.

4 Discussion

4.1 Employment Rate in Ontario

In Figure 1, we observe the employment rate fluctuating from around 57.9%-60.5%, with regular seasonal fluctuations that tend to be higher in the summer months, and lower in the winter months. At the 2018 minimum wage change, there is a sharp drop in employment, to 57.9% at which is the lowest point of employment rate in the series. This drop is substantially larger than the small decrease observed after the 2017 minimum wage change. After the 2018 change, the employment rate recovers relatively quickly, before experiencing another drop, suggesting the presence of some seasonality. This could imply that the initial drop may have overstated the effect of the 2018 minimum wage change.

4.2 Employment Rate by Age

Figure 2 highlights different employment patterns depending on the age group. The two youth groups, which are ages 15-19 and 20-24 show similar cycles and tend to fluctuate together throughout the entire series. The 15-19 group is the most volatile, fluctuating at around 38%-53%. The 20-24 group is also volatile, but less so, fluctuating at around 62%-74%. The comparison group, aged 30-49 represent what most would consider the prime working age group. The employment rate for this group is noticeably flatter, and much less volatile (81%-83%), characterized by more experience, stable income, and less exposure to minimum wages. An interesting observation is that the January 2018 policy change is not associated with the largest fluctuations. Rather, youth employment fluctuation tends to be seasonal, with big increases in the summer months, rather than after the minimum wage changes.

4.3 Youth Employment Rate by Industry

Figure 3 provides four comparison groups, where two are regarded as youth-dominated, and the other two are not. This is evident in the graph, as a larger share of youth are concentrated in retail (13.5%) and food services (11%), rather than health care (4%) and professional (2.5%) sectors. It also follows that retail and food service employment is slightly more volatile than the other two, even among the younger population. After the 2018 policy change in particular, retail has the largest visible drop, while the other three remain stable.

4.4 Youth Employment by Full-Time and Part-Time

In Figure 4, we compare youth employment of full-time and part-time status, inclusive of all industries, with the full-time rate fluctuating around 58%, and part-time fluctuating around 11%. This is likely due to our youth measure pooling teenagers of 15-19 with young adults of 20-24, who are mostly out of school and/or working full-time. The January 2018 dip in the full-time trend exists, but is small. The part-time trend on the other hand, is essentially flat across the entire three years, with no significant change in January 2018. The full-time and part-time lines generally move as mirror images of each other throughout. This is a compelling observation as it indicates a fixed underlying employment rate that's just being reallocated seasonally between full-time and part-time status, likely due to factors like school.

4.5 Youth Employment in Retail/Food Service by Full-Time and Part-Time

Figure 5 captures a youth employment rate of around 65%-85% for retail and food service workers who work part-time, and 20%-40% for retail and food service workers who work full-time. The seasonality posits that part-time work increases in the winter months and decreases in the summer months, while full-time work presents the opposite pattern. This is consistent with the schooling effect that youth are subject to, as discussed above. Although there is seasonal volatility, there is no distinct change following the 2018 wage increase. This suggests that students are the main demographic in these industries, and strong seasonal patterns in their employment obscure any clear minimum wage effects.

4.6 Employment Rate by Education Level

Figure 6 demonstrates a clear consistent gap between higher and lower education levels, with the postsecondary and above trend fluctuating around 68%, and the high school and below trend fluctuating around 44%. The latter is noticeably more volatile, which is expected given it's a smaller, more precarious segment of the labour market. Both trends decrease very slightly in January 2018, but it's difficult to isolate a policy change from regular volatility, especially since similar dips can be seen later on, for instance, mid 2018 and mid 2019. The postsecondary line is comparatively flat, which is expected if this group is largely isolated from minimum wage level jobs regardless of age.

4.7 Conclusion

In summary, none of the six figures exhibit a disruption in January 2018 that is distinguishable, with certainty, from the recurring seasonal variation in the sample. Youth retail employment shows a modest, short-lived decline following the increase, but this is not reflected in accommodation and food services, nor by any shift in the full-time/part-time composition of youth work. The visual evidence, can therefore only be deemed inconclusive, because a genuine effect may be obscured by seasonality that the time series in itself cannot net out.

5 Stata Code

```
1
2 clear
3
4
5 use "/Users/averybennett/Desktop/ECON 422/LFS (Econ 422 Exercise 1).dta"
6
7 log using "/Users/averybennett/Desktop/ECON 422/ECON 422.log", replace
8
9 /*
10  - Looking at labour force status, age, industry, and time frame (survey month and survey
    year)
11  - Basic visualizing of the data, for instance look at tabulations (tab command) and
    distributions for wages (sum, detail) of survey month and survey year and some of the
    other key variables.
12 */
13
14 tab SURVMNTH
15 tab SURVYEAR
16 tab LFSSTAT
17 tab NAICS_21
18 tab EDUC
19 tab AGE_12
20 sum HRLYEARN, detail
21
22
23 rename SURVYEAR year
24 rename SURVMNTH month
25 rename AGE_12 age
26 rename PROV province
```



```

27
28 keep if province == 35
29
30 gen employed = .
31 replace employed = 1 if LFSSTAT<=2
32 replace employed = 0 if LFSSTAT>=3
33
34 gen youth = age <= 2
35
36 gen date = ym(year, month)
37 format date %tm
38
39 keep if date >= ym(2017,1) & date <= ym(2019,12)
40
41 tab youth if employed == 1 & NAICS_21 == 10
42 tab youth if employed == 1 & NAICS_21 == 19
43 tab youth if employed == 1 & NAICS_21 == 17
44 tab youth if employed == 1 & NAICS_21 == 14
45
46 /*
47  - Creating a monthly time series of the employment rate using the LFS data.
48  */
49
50 rename SURVYEAR year
51 rename SURVMNTH month
52 rename PROV province
53 keep if province == 35
54
55 gen date = ym(year, month)
56 format date %tm
57 keep if date >= ym(2017,1) & date <= ym(2019,12)
58
59 egen emp_rate = mean(employed), by(date)
60 replace emp_rate = emp_rate*100
61
62 twoway (line emp_rate date, sort), xtitle("Month") ytitle("Employment Rate (%)") title("
    Ontario Employment Rate Monthly Time Series")
63
64 /*
65  - Subsamples of the employment rate.
66  - By age groups.
67  */
68

```

```

69 clear
70
71 use "/Users/averybennett/Desktop/ECON 422/LFS (Econ 422 Exercise 1).dta"
72
73 rename SURVYEAR year
74 rename SURVMNTH month
75 rename AGE_12 age
76 rename PROV province
77 keep if province == 35
78
79 gen employed = .
80 replace employed = 1 if LFSSTAT<=2
81 replace employed = 0 if LFSSTAT>=3
82
83 gen date = ym(year, month)
84 format date %tm
85 keep if date >= ym(2017,1) & date <= ym(2019,12)
86
87 egen emp_rate1 = mean(employed) if age == 1, by(date)
88 replace emp_rate1 = emp_rate1*100
89
90 egen emp_rate2 = mean(employed) if age == 2, by(date)
91 replace emp_rate2 = emp_rate2*100
92
93 egen emp_rate3 = mean(employed) if age >= 4 & age <= 7, by(date)
94 replace emp_rate3 = emp_rate3*100
95
96 twoway (line emp_rate1 date, sort) (line emp_rate2 date, sort) (line emp_rate3 date, sort
      ), xtitle("Month") ytitle("Employment Rate (%)") legend(position(6) label(1 "Ages
      15-19") label(2 "Ages 20-24") label(3 "Ages 30-49")) title("Ontario Employment Rate
      by Age Group")
97
98 /*
99 - Retail trade and Accommodation and food services industries are youth dominated, while
      Healthcare and social assistance and Professional, scientific, and technical services
      are not.
100 */
101
102 clear
103
104 use "/Users/averybennett/Desktop/ECON 422/LFS (Econ 422 Exercise 1).dta"
105
106 rename SURVYEAR year

```

```

107 rename SURVMNTH month
108 rename AGE_12 age
109 rename PROV province
110 keep if province == 35
111 keep if age <=2
112
113 gen employed = .
114 replace employed = 1 if LFSSTAT<=2
115 replace employed = 0 if LFSSTAT>=3
116
117 gen date = ym(year, month)
118 format date %tm
119 keep if date >= ym(2017,1) & date <= ym(2019,12)
120
121 gen retail_youth = (employed == 1 & NAICS_21 == 10)
122 egen retail_rate = mean(retail_youth), by(date)
123
124 gen food_youth = (employed == 1 & NAICS_21 == 19)
125 egen food_rate = mean(food_youth), by(date)
126
127 gen health_youth = (employed == 1 & NAICS_21 == 17)
128 egen health_rate = mean(health_youth), by(date)
129
130 gen pro_youth = (employed == 1 & NAICS_21 == 14)
131 egen pro_rate = mean(pro_youth), by(date)
132
133 replace retail_rate = retail_rate * 100
134 replace food_rate = food_rate * 100
135 replace health_rate = health_rate * 100
136 replace pro_rate = pro_rate * 100
137
138 twoway (line retail_rate date, sort) (line food_rate date, sort) (line health_rate date,
        sort) (line pro_rate date, sort), xtitle("Month") ytitle("Percentage of Youth
        Population") legend(position(6) label(1 "Retail Trade") label(2 "Accommodation and
        Food Services") label(3 "Health Care and Social Assistance") label(4 "Professional,
        Scientific and Technical Services")) title("Ontario Youth Employment by Industry")
139
140 /*
141 Full-time and part-time employment among youth.
142 */
143
144 clear
145

```

```

146 use "/Users/averybennett/Desktop/ECON 422/LFS (Econ 422 Exercise 1).dta"
147
148 rename SURVYEAR year
149 rename SURVMNTH month
150 rename AGE_12 age
151 rename PROV province
152
153 keep if province == 35
154 keep if age <= 2
155
156 gen date = ym(year, month)
157 format date %tm
158 keep if date >= ym(2017,1) & date <= ym(2019,12)
159
160 gen employed = LFSSTAT <= 2
161 gen ft_youth = (employed == 1 & FTPTMAIN == 1) * 100
162 gen pt_youth = (employed == 1 & FTPTMAIN == 2) * 100
163
164 egen youth_fulltime_rate = mean(ft_youth), by(date)
165 egen youth_parttime_rate = mean(pt_youth), by(date)
166
167 twoway (line youth_fulltime_rate date, sort) (line youth_parttime_rate date, sort),
        xtitle("Month") ytitle("Percentage of Youth Population") legend(position(6) label(1 "
        Full-Time") label(2 "Part-Time")) title("Ontario Youth Employment by Full-Time and
        Part-Time Status")
168
169 /*
170 Full vs part time just retail/food
171 */
172 clear
173
174 use "/Users/averybennett/Desktop/ECON 422/LFS (Econ 422 Exercise 1).dta"
175
176 rename SURVYEAR year
177 rename SURVMNTH month
178 rename PROV province
179 keep if province == 35
180
181 gen employed = .
182 replace employed = 1 if LFSSTAT<=2
183 replace employed = 0 if LFSSTAT>=3
184
185 gen date = ym(year, month)

```

```

186 format date %tm
187 keep if date >= ym(2017,1) & date <= ym(2019,12)
188
189 rename AGE_12 age
190 keep if age <= 2
191
192 keep if employed == 1 & (NAICS_21 == 10 | NAICS_21 == 19)
193
194 gen ft = (FTPTMAIN == 1) * 100
195 egen fulltime_rate = mean(ft), by(date)
196
197 gen pt = (FTPTMAIN == 2) * 100
198 egen parttime_rate = mean(pt), by(date)
199
200 twoway (line fulltime_rate date, sort) (line parttime_rate date, sort), xtitle("Month")
      ytitle("Percentage of Youth Employed in Retail/Food") legend(position(6) label(1 "
      Full-Time") label(2 "Part-Time")) title("Full-Time vs. Part-Time Status for Ontario
      Youth in Retail/Food Services")
201
202 /*
203 Education level employment
204 */
205 clear
206
207 use "/Users/averybennett/Desktop/ECON 422/LFS (Econ 422 Exercise 1).dta"
208
209 rename SURVYEAR year
210 rename SURVMNTH month
211 rename PROV province
212 keep if province == 35
213
214 gen employed = .
215 replace employed = 1 if LFSSTAT<=2
216 replace employed = 0 if LFSSTAT>=3
217
218 gen date = ym(year, month)
219 format date %tm
220 keep if date >= ym(2017,1) & date <= ym(2019,12)
221
222 gen low_educ = EDUC <= 2
223
224 egen low_educ_rate = mean(employed) if low_educ == 1, by(date)
225 replace low_educ_rate = low_educ_rate * 100

```

```

226
227 egen high_educ_rate = mean(employed) if low_educ == 0, by(date)
228 replace high_educ_rate = high_educ_rate * 100
229
230 twoway (line low_educ_rate date, sort) (line high_educ_rate date, sort), xtitle("Month")
      ytitle("Employment Rate (%)") legend(position(6) label(1 "High School or Below") label
      (2 "Postsecondary and Above")) title("Ontario Employment Rate by Education Level")
231
232 log close

```

6 Stata Output

```
tab SURVMNTH
```

Survey			
month	Freq.	Percent	Cum.
-----+-----			
January	140,094	8.55	8.55
2	139,557	8.52	17.07
3	138,143	8.43	25.50
4	137,318	8.38	33.89
5	136,752	8.35	42.23
6	136,873	8.36	50.59
7	135,233	8.26	58.85
8	135,160	8.25	67.10
9	135,585	8.28	75.37
10	135,136	8.25	83.62
11	134,566	8.21	91.84
12	133,726	8.16	100.00
-----+-----			
Total	1,638,143	100.00	

```
. tab SURVYEAR
```

Survey year	Freq.	Percent	Cum.
-----+-----			
2016	336,569	20.55	20.55

2017		334,474	20.42	40.96
2018		333,777	20.38	61.34
2019		338,360	20.66	81.99
2020		294,963	18.01	100.00
-----+-----				
Total		1,638,143	100.00	

. tab LFSSTAT

Labour force status		Freq.	Percent	Cum.
-----+-----				
Employed, at work		872,735	53.28	53.28
Employed, absent from work		83,712	5.11	58.39
Unemployed		50,622	3.09	61.48
Not in labour force		505,523	30.86	92.34
5		585	0.04	92.37
6		124,966	7.63	100.00
-----+-----				
Total		1,638,143	100.00	

. tab NAICS_21

Industry of main job		Freq.	Percent	Cum.
-----+-----				
Agriculture		13,130	1.54	1.54
Forestry and logging and support activi		1,959	0.23	1.77
Fishing, hunting and trapping		132	0.02	1.79
Mining, quarrying, and oil and gas extr		8,278	0.97	2.76
Utilities		7,038	0.83	3.59
Construction		65,986	7.75	11.34
Manufacturing - durable goods		53,186	6.25	17.59
Manufacturing - non-durable goods		36,841	4.33	21.92
Wholesale trade		26,441	3.11	25.03
Retail trade		96,837	11.38	36.41
Transportation and warehousing		41,109	4.83	41.24
Finance and insurance		36,088	4.24	45.48

Real estate and rental and leasing	15,300	1.80	47.28
Professional, scientific and technical	61,180	7.19	54.47
Business, building and other support se	35,063	4.12	58.59
Educational services	68,594	8.06	66.65
Health care and social assistance	108,715	12.78	79.43
Information, culture and recreation	37,102	4.36	83.79
Accommodation and food services	57,496	6.76	90.54
Other services (except public administr	33,964	3.99	94.54
Public administration	46,504	5.46	100.00
-----+-----			
Total	850,943	100.00	

. tab EDUC

Highest educational attainment	Freq.	Percent	Cum.
-----+-----			
0 to 8 years	53,219	4.09	4.09
Some high school	149,719	11.50	15.59
High school graduate	281,337	21.62	37.21
Some postsecondary	82,374	6.33	43.54
Postsecondary certificate or diploma	417,359	32.07	75.60
Bachelor's degree	208,695	16.03	91.64
Above bachelor's degree	108,871	8.36	100.00
-----+-----			
Total	1,301,574	100.00	

. tab AGE_12

Five-year age			
group of			
respondent	Freq.	Percent	Cum.
-----+-----			
15 to 19 years	111,673	6.82	6.82
20 to 24 years	109,880	6.71	13.52
25 to 29 years	115,648	7.06	20.58
30 to 34 years	119,185	7.28	27.86

35 to 39 years		120,445	7.35	35.21
40 to 44 years		120,138	7.33	42.55
45 to 49 years		128,229	7.83	50.37
50 to 54 years		147,340	8.99	59.37
55 to 59 years		155,077	9.47	68.83
60 to 64 years		142,734	8.71	77.55
65 to 69 years		121,462	7.41	84.96
70 and over		246,332	15.04	100.00
-----+-----				
Total		1,638,143	100.00	

. sum HRLYEARN, detail

Usual hourly wages, employees only

Percentiles		Smallest		
1\%	10.33	2.02		
5\%	11.7	2.02		
10\%	13.74	2.4	Obs	813,492
25\%	16.05	2.49	Sum of wgt.	813,492
50\%	24		Mean	27.4145
		Largest	Std. dev.	14.36194
75\%	35	153.85		
90\%	47	155.13	Variance	206.2654
95\%	54.87	161.54	Skewness	1.42517
99\%	75	192.31	Kurtosis	5.759789

. tab youth if employed == 1 & NAICS_21 == 10

youth		Freq.	Percent	Cum.
-----+-----				
0		49,758	73.99	73.99
1		17,493	26.01	100.00
-----+-----				

Total		67,251	100.00
-------	--	--------	--------

```
. tab youth if employed == 1 & NAICS_21 == 19
```

youth		Freq.	Percent	Cum.
-----+-----				
0		22,865	59.35	59.35
1		15,659	40.65	100.00
-----+-----				
Total		38,524	100.00	

```
. tab youth if employed == 1 & NAICS_21 == 17
```

youth		Freq.	Percent	Cum.
-----+-----				
0		73,301	92.95	92.95
1		5,563	7.05	100.00
-----+-----				
Total		78,864	100.00	

```
. tab youth if employed == 1 & NAICS_21 == 14
```

youth		Freq.	Percent	Cum.
-----+-----				
0		40,611	93.61	93.61
1		2,772	6.39	100.00
-----+-----				
Total		43,383	100.00	

```
.
```

7 References

Government of Canada, Statistics Canada. (2017, April 7). *Labour Market Indicators*. Statistics Canada. <https://www150.statcan.gc.ca/n1/pub/71-607-x/71-607-x-2017001-eng.htm>

General Hourly Minimum Wage Rates in Canada since 1965. Government of Canada. (n.d.). <https://minwage-salairemin.service.canada.ca/en/since1965.html>

I used Claude AI for the following:

- Formatting the date for my twoway graphs.
- For how to use "or" and "if" to code some of my variables.
- Putting my legend below the graph.
- I used it to check the syntax of my code, particularly for gen, egen, and replace. I sometimes got confused for the order, and I wanted to make sure my arguments were doing what I intended.

This was done alongside the following sources:

- <https://www.stata.com/manuals/g-2graphtwoway.pdf#g-2graphtwoway>
- <https://www.stata.com/manuals/ddatetime.pdf>
- <https://stats.oarc.ucla.edu/stata/modules/using-if-with-stata-commands/>
- https://www.stata.com/manuals/g-3legend_options.pdf